

Title: LLM-Based Evaluation vs Non-LLM-Based Evaluation

Subtitle: A Comparative Analysis for AI Testing

Presented by: [Your Name] – SDET Specialist | [Date]

Slide 2: Agenda

- What is Evaluation in LLM Testing?
 - Evaluation Categories
 - LLM-Based Evaluation Explained
 - Non-LLM-Based Evaluation Explained
 - Key Differences: LLM vs Non-LLM Evaluation
 - Pros & Cons
 - Use Case Scenarios
 - Recommendation Matrix
 - Conclusion
-

Slide 3: What is Evaluation in LLM Testing?

Definition:

Evaluation is the process of measuring the quality, relevance, accuracy, and usefulness of a language model's response using defined metrics.

Why it matters:

- Ensures real-world performance
- Reduces hallucination risks
- Tracks improvement over time

- Informs fine-tuning

Slide 4: Categories of Evaluation

Category	Description
LLM-Based Evaluation	Uses another language model to judge output
Non-LLM Evaluation	Uses rule-based or statistical techniques (no LLM)
Human Evaluation	Manual scoring by humans based on rubric

Slide 5: LLM-Based Evaluation

Definition:

Uses a language model (e.g., GPT-4) to assess responses based on rubrics or qualitative criteria.

Example:

Prompt: *"Rate this answer from 1-5 for helpfulness."*

Model: *"Score: 4. The answer was mostly relevant and coherent."*

Common Tools:

- DeepEval
 - OpenAI Function Calling
 - LLM-As-A-Judge (Chain-of-Thought + rubric)
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Slide 6: Non-LLM Based Evaluation

Definition:

Uses traditional or algorithmic methods to evaluate output without relying on another LLM.

Examples:

- **BLEU / ROUGE:** Compare n-gram overlap with reference
- **Embedding Similarity:** Cosine similarity between sentence vectors
- **Edit Distance / Jaccard:** Token-level or set-level similarity

Common Tools:

- RAGAS
- scikit-learn metrics
- NLTK / spaCy

Slide 7: Comparison Table

Feature	LLM-Based Evaluation	Non-LLM-Based Evaluation
Dependency	Requires another LLM	No LLM required
Interpretability	Moderate (may need CoT)	High (based on algorithm)
Customization	Highly customizable (rubrics)	Limited by metric type
Cost (tokens)	High	Low
Speed	Slower (LLM latency)	Faster (local computation)
Bias Potential	Can inherit LLM bias	More objective
Use Case Fit	Subjective/complex responses	Factual/objective comparison

Slide 8: Pros & Cons

LLM-Based Evaluation

- ✓ Understands nuance & context
- ✓ Scales human-like judgment
- ✗ Expensive (token usage)
- ✗ Requires prompt engineering

Non-LLM Evaluation

- ✓ Fast and cheap
 - ✓ Transparent metrics
 - ✗ Poor with open-ended answers
 - ✗ No understanding of intent
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Slide 9: Use Case Scenarios

Use Case	Recommended Approach	Why
Factual Q&A Chatbot	Non-LLM Evaluation	Easy to compare with references
Creative Writing Assistant	LLM-Based Evaluation	Needs contextual and style judgment
RAG (Retrieval-Augmented) QA	Hybrid	Use embedding + LLM reasoning
Customer Support Bot	LLM-Based	Needs tone and empathy evaluation

Slide 10: Recommendation Matrix

Criteria	LLM-Based Eval	Non-LLM Eval
Cost Sensitive	✗	✓
Real-time Scoring	✗	✓
Subjective Judgments	✓	✗
Ground Truth Present	✗	✓
Complex Reasoning	✓	✗

Slide 11: Conclusion

- **LLM-Based** evaluation brings **human-like reasoning**, ideal for **subjective, nuanced outputs**.
- **Non-LLM-Based** metrics are **fast, cheap, and interpretable**, but lack deeper understanding.
- Best results often come from a **hybrid approach** depending on your application type.

 Slide 12: Bonus – Frameworks You Can Use

Framework	LLM-Based	Non-LLM-Based	Use Case
RAGAS	✓	✓	RAG Evaluation
DeepEval	✓	✓	Multi-metric eval, customizable
LangSmith	✓	✗	Tracing and LLM evals
Promptfoo	✓	✓	Prompt optimization/testing

Would you like me to export this into a `.pptx` file format or design a sample slide layout using visuals and graphs?

You said:

NOt fit Need to refine and give me difference and with some good example

ChatGPT said:

Great — let's **refine** the comparison and present the **differences between LLM-Based and Non-LLM-Based Evaluation** with **strong examples**, and make it **presentation-ready** for a professional audience (QA leads, engineers, CTOs).

 Slide Title: LLM-Based Evaluation vs Non-LLM-Based Evaluation

A Clear Comparison with Real-World Examples

Slide 1: Evaluation Techniques in LLM Testing

Type of Evaluation	Description
LLM-Based Evaluation	Uses a language model (LLM) to assess output quality based on rubrics
Non-LLM-Based Evaluation	Uses fixed algorithms (e.g., BLEU, ROUGE, cosine similarity) to compare with reference

Slide 2: Key Differences at a Glance

Criteria	LLM-Based Evaluation	Non-LLM-Based Evaluation
Uses another LLM?	✅ Yes (e.g., GPT-4 judges GPT-4 output)	❌ No LLM involved
Subjectivity support	✅ Handles nuance, tone, and style	❌ Only measures surface-level similarity
Speed & Cost	❌ Slower, high token cost	✅ Fast, cheap, scalable
Interpretability	❌ Sometimes opaque (depends on LLM's reasoning)	✅ Transparent, deterministic
Best Fit For	Complex tasks, open-ended answers	Simple Q&A, fact-matching
Bias Risk	⚠️ Can reflect LLM's internal bias	✅ Rule-based, less prone to bias

Slide 3: Example – Chatbot Answer Evaluation

Scenario: Evaluating this chatbot answer to “What causes rain?”

Model's Answer: *“Rain is caused by water vapor condensing into droplets in clouds.”*

With Non-LLM Evaluation (Embedding Similarity)

- **Reference Answer:** “Rain forms when water vapor in the air condenses into clouds and falls as precipitation.”
- **Result:** Cosine similarity score = **0.82**
- **Limitation:** Doesn't judge clarity or completeness

With LLM-Based Evaluation (LLM as Judge)

- **Prompt:** “Rate this answer for helpfulness, completeness, and accuracy (1–5).”
 - **LLM Reply:** “Score: 5 – The explanation is accurate and well-stated.”
 - **Advantage:** Captures judgment beyond surface matching
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Slide 4: Example – Creative Writing Response

Task: Generate a poetic summary of the day’s weather

Candidate Output: “The sky weeps softly as clouds embrace the land.”

Non-LLM Evaluation:

- ROUGE score = **Low** (doesn't match any reference text)
- Verdict:  Poor score

LLM-Based Evaluation:

- GPT-4 rubric: “Creativity: 5, Language: 5, Relevance: 4 → Final Score: 4.7”
 - Verdict:  Excellent poetic style and relevance recognized
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Slide 5: When to Use What?

Scenario	Recommended Evaluation Type	Reason
Factual Answer Matching (e.g., RAG QA)	 Non-LLM-Based (e.g., RAGAS)	Objective comparison against source/reference
Summarization/Paraphrasing	 LLM-Based	Style, flow, and compression quality matter
Safety/Toxicity/Tone Checks	 LLM-Based	Nuanced, subjective aspects need reasoning

Multi-turn Conversations	✅ LLM-Based	Tracks coherence and context handling
Search Relevance Scoring	✅ Non-LLM-Based (embedding)	Ranking based on vector similarity

Slide 6: Summary – Which One to Choose?

Need	Choose
Low cost, high speed	Non-LLM-Based
Evaluating factual accuracy	Non-LLM-Based (BLEU/Recall)
Evaluating subjective tasks	LLM-Based
Fine-tuning via rubric	LLM-Based
Production-level QA bots	Hybrid (LLM + Traditional)

Bonus Slide: Tooling Landscape

Tool/Framework	LLM-Based	Non-LLM-Based	Notes
RAGAS	✅	✅	Great for evaluating RAG pipelines
DeepEval	✅	✅	Customizable, supports CoT + rules
Promptfoo	✅	✅	Local + hosted evaluation
OpenAI Eval	✅	❌	LLM-as-a-Judge testing

Would you like a **PowerPoint file (.pptx)** with design, icons, and slide formatting based on this content? Or should I deliver this in **Google Slides / PDF format**?

You said:

LLM Based metrics Non LLM based metrics definition and disadvantage

ChatGPT said:

Here's a **clear and professional slide-style explanation** of:

- LLM-Based Metrics
 - Non-LLM-Based Metrics
 - Their **definitions and disadvantages**
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Slide Title: LLM-Based Metrics vs Non-LLM-Based Metrics

LLM-Based Metrics

Definition:

Metrics generated by using a large language model (LLM) to evaluate another model's output based on quality, coherence, helpfulness, tone, etc.

LLMs act as **judges** based on rubrics, prompts, or chain-of-thought reasoning.

Example:

Prompt: "Rate the answer on helpfulness (1–5)"

LLM Response: "Score = 4. The answer is mostly correct and clear."

Used In:

- Summarization
- Open-ended QA
- Generative dialog
- Subjective task scoring

Disadvantages:

-  **Costly**: Consumes tokens → higher inference cost
-  **Latency**: Slower due to LLM call overhead
-  **Biased**: LLMs may inject subjective bias or hallucinate judgments

- **✗ Opaque:** Reasoning can be hard to trace or debug
 - **✗ Inconsistent:** Same prompt may give different scores without prompt engineering
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Non-LLM-Based Metrics

Definition:

Rule-based or algorithmic metrics that measure similarity between model outputs and reference texts, without using any LLM.

Examples of Metrics:

- **BLEU / ROUGE:** n-gram overlap
- **Cosine Similarity:** Embedding distance between output and reference
- **Edit Distance / Levenshtein:** Token-level differences
- **Exact Match (EM):** Binary 0 or 1 scoring

Used In:

- RAG evaluation
- Fact-based QA
- Machine translation
- Named Entity Matching

Disadvantages:

- **✗ Surface-Level:** Doesn't understand meaning or context
- **✗ Brittle:** Penalizes paraphrasing even if the answer is correct
- **✗ Not Suitable for Open-Ended Tasks:** Can't judge creativity, tone, nuance
- **✗ Insensitive to Grammar & Style:** Doesn't recognize quality of expression

- **✗ Reference Dependent:** Requires gold standard/reference outputs

✓ Summary Table

Aspect	LLM-Based Metrics	Non-LLM-Based Metrics
Uses LLM?	✓ Yes	✗ No
Understands meaning?	✓ Yes	✗ Only surface similarity
Cost	✗ High (token + latency)	✓ Low (local, fast)
Bias risk	✗ Can be biased	✓ Deterministic
Suited For	Subjective, generative tasks	Factual, deterministic outputs
Examples	GPT-4 scoring, DeepEval	BLEU, ROUGE, cosine similarity, EM

Would you like me to generate this content as **visually formatted PowerPoint slides** with icons and diagrams?

You said:

with some example

ChatGPT said:

Absolutely — here's a **refined, professional breakdown** of **LLM-Based Metrics vs Non-LLM-Based Metrics** with **clear definitions, real examples, and disadvantages**, formatted as **presentation slides**.

🎯 Slide 1: Title

LLM-Based Metrics vs Non-LLM-Based Metrics

Evaluating Language Model Output: Techniques, Examples & Pitfalls

Slide 2: What Are LLM-Based Metrics?

Definition:

LLM-based metrics use another **language model** (like GPT-4) to **score** or **judge** the output based on prompts, rubrics, or chain-of-thought reasoning.

Example:

Task: “Summarize: The Earth revolves around the Sun due to gravity.”

Generated Summary: “Gravity causes Earth to orbit the Sun.”

LLM Metric Prompt: “Rate this summary on accuracy and conciseness (1–5)”

LLM Judgment: “Score: 5 – Accurate and concise summary.”

Use Cases:

- Summarization
 - Chatbot helpfulness
 - Tone and empathy evaluation
 - Answer quality in open-ended QA
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Slide 3: Disadvantages of LLM-Based Metrics

Issue	Description
 High Cost	Each evaluation uses tokens → expensive at scale
 Slow Latency	Requires real-time LLM call → not real-time friendly
 Inconsistent	May produce different scores for same output unless prompts are engineered well
 Bias Risk	Can inherit LLM’s training biases or hallucinations
 Low Transparency	It’s hard to explain <i>why</i> an LLM gave a specific score



Slide 4: What Are Non-LLM-Based Metrics?

Definition:

Non-LLM-based metrics are rule-based or statistical metrics that **compare model output to a reference** without requiring another language model.

Example:

Reference Answer: “Water vapor condenses into droplets forming clouds.”

Model Output: “Rain happens when vapor condenses into water droplets.”

Metric Used: Cosine Similarity on Sentence Embeddings

Score: 0.87 (high semantic similarity)

Common Metrics:

- BLEU: n-gram overlap
- ROUGE: recall-based text comparison
- Cosine similarity: embedding-based
- Exact Match / Levenshtein distance

 Slide 5: Disadvantages of Non-LLM-Based Metrics

Issue		Description
 Lack of Understanding	Doesn't capture meaning or context	
 Rigid Matching	Penalizes valid paraphrases	
 Can't Handle Subjectivity	Doesn't assess tone, empathy, or style	
 Needs Ground Truth	Reference answers must be available	
 No Creativity Scoring	Can't score poems, jokes, summaries properly	

 Slide 6: Side-by-Side Comparison with Example

Metric Type	Evaluation Method	Example Task	Output	Score	Notes
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Non-LLM (ROUGE)	Text overlap with reference	Summarization	“Gravity makes Earth orbit.”	0.41	Low overlap despite correctness
Non-LLM (Cosine)	Embedding similarity	QA Answer	“It’s gravity that pulls Earth.”	0.89	Recognizes meaning
LLM-Based	GPT-4 rubric scoring	Summarization / QA	“Gravity causes orbit.”	5/5	Judges clarity, correctness
LLM-Based	Chain-of-Thought explanation	Chatbot empathy check	“I’m sorry you’re feeling that way.”	“4 – Compassionate but short”	Context-aware judgment

✓ Slide 7: Summary – Which Should You Use?

If You Need...	Use This Type	Why?
Fast, cheap, repeatable scoring	Non-LLM-Based	Deterministic and scalable
Subjective or nuanced judgment	LLM-Based	Handles tone, clarity, intent
RAG system evaluation (e.g., context recall)	Hybrid (e.g., RAGAS)	Embeddings + GPT rubrics
Fact matching in QA	Non-LLM	Better for objective tasks
Human-like feedback	LLM-Based	Mimics real user perception